

A Potent Fluorescent Reversible-Covalent Inhibitor of Cardiac Muscle Contraction

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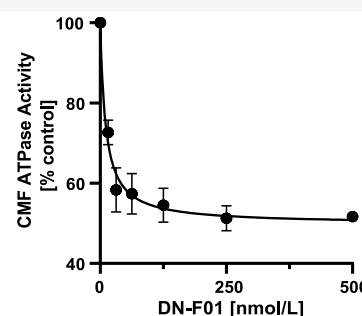
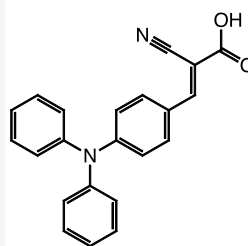
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ABSTRACT: Compounds that directly modulate the response of the cardiac sarcomere have potential in the treatment of cardiac disease. While a number of sarcomere activators have been discovered and extensively studied, very few inhibitors have been identified. We report a potent cardiac sarcomere inhibitor, DN-F01, targeting the cardiac muscle thin filament protein troponin complex. Functional studies show that DN-F01 has a strong inhibitory calcium-dependent effect on cardiac myofibrillar ATPase activity with an IC_{50} value of 11 ± 4 nmol/L. DN-F01 is shown to bind to a cardiac troponin C–troponin I chimera (cChimera) with a K_D of ~ 50 nM using fluorescence spectroscopy, indicating that troponin is the likely target for DN-F01. NMR titrations of DN-F01 to C35S and A-Cys cChimera show covalent and noncovalent

KEYWORDS: cardiac, troponin, sarcomere, inhibitor



binding of DN-F01 bound to the calcium-saturated cChimera.

Both systolic and diastolic dysfunction can lead to heart failure. Systolic dysfunction occurs when the heart muscle is weak and dilated, losing its ability to contract; while diastolic dysfunction is caused by thick and stiff heart muscle, impairing the relaxation process. Common therapeutics that are available increase or decrease cardiac muscle contractility by regulating the calcium-based signaling pathway and alter the calcium transients, but this can also affect other signaling systems.¹ Compounds that directly and specifically modify the response of the cardiac sarcomere would be ideal for the treatment of cardiac disease.

Contraction of the heart is regulated by calcium binding to cardiac troponin (cTn) in the thin filament. cTn has three subunits: the calcium-binding troponin C (cTnC), the inhibitory troponin I (cTnI), and the tropomyosin-binding troponin T (cTnT).² When calcium binds to the N-terminal domain of cTnC (cNTnC), it undergoes a structural change that promotes the binding of the cTnI switch region.³ This interaction pulls the inhibitory region of cTnI off actin, shifting tropomyosin to expose the myosin binding sites on actin, which initiates contraction.⁴ Since the calcium dependent trigger is the association of cNTnC and cTnI, this interface is the logical target for the development of direct cardiac sarcomere modulating compounds.

Several small molecules are known to bind to the interface of cNTnC and cTnI and alter the calcium sensitivity of the sarcomere, such as bepridil,⁵ pimobendan,⁶ levosimendan,⁷ dfbp-o,⁸ and 3Cl-DPA.⁹ The most widely studied activator bound to this interface is levosimendan, which has marketing authorization in over 60 countries worldwide to treat acute

heart failure.^{10,11} Recent studies show that levosimendan forms a reversible-covalent thioimide bond with Cys 84 of cNTnC,^{12,13} indicating that this unique cysteine has the potential for developing new compounds. While equally important as activators, very few inhibitors have been identified. N-(6-Aminoethyl)-5-chloro-1-naphthalenesulfonamide (W7) is a well-studied direct sarcomere inhibitor that binds to this interface,^{14,15} and it decreased the binding of cTnI to cNTnC by about 10-fold,¹⁶ possibly by steric clashing and electrostatic repulsion between W7 and cTnI.¹⁷ Another inhibitor is epigallocatechin gallate (EGCG), but it binds to the structural C-domain of cTnC (cTnC).¹⁸

In this Note, we report a potent inhibitor, DN-F01 (4-(diphenylamino)phenylcyanoacrylic acid). DN-F01 was initially used as a fluorescent dye, with the triphenylamine donor and the cyanoacetic acid acceptor.¹⁹ DN-F01 shares structural similarities to the principal compounds we have studied for their sarcomere activating potential: diphenylamine (DPA)-containing compounds and the notable activator levosimendan (Figure 1). Compounds based on diphenylamine (DPA) show moderate binding affinity to a cNTnC–cTnI chimera ($K_D \sim 10$ – $120 \mu M$).²⁰ NMR solution structures indicate that 3-

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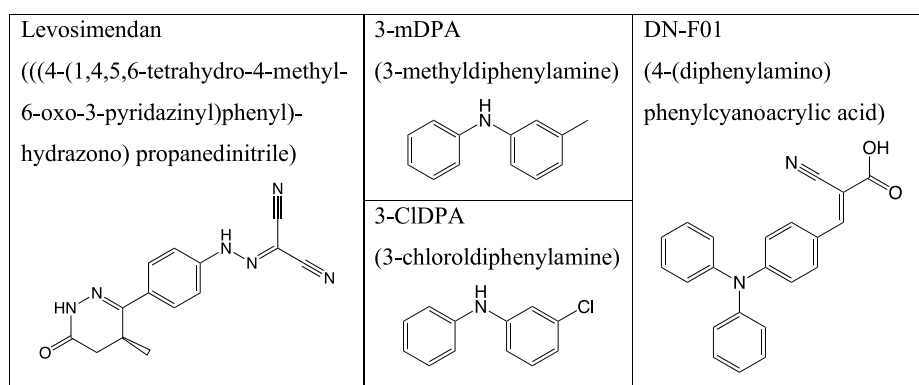


Figure 1. Chemical structures of levosimendan, 3-mDPA, 3-CIDPA, and DN-F01.

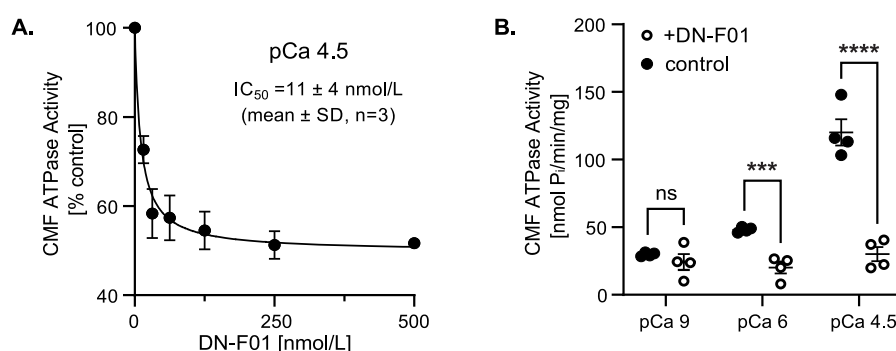


Figure 2. (A) Dose–response curve for the effect of DN-F01 on cardiac myofibrillar ATPase activity at full calcium activation (pCa 4.5). (B) Effect of DN-F01 on the CMF ATPase activity at low (pCa 9), intermediate (pCa 6), and high (pCa 4.5) calcium concentrations. Means $\pm$ SEM, $n = 3$ –4. Statistical significance of differences between groups was assessed with a two-way ANOVA followed by Sidak's multiple comparison test: ns, not significant; *** $P < 0.001$; **** $P < 0.0001$.

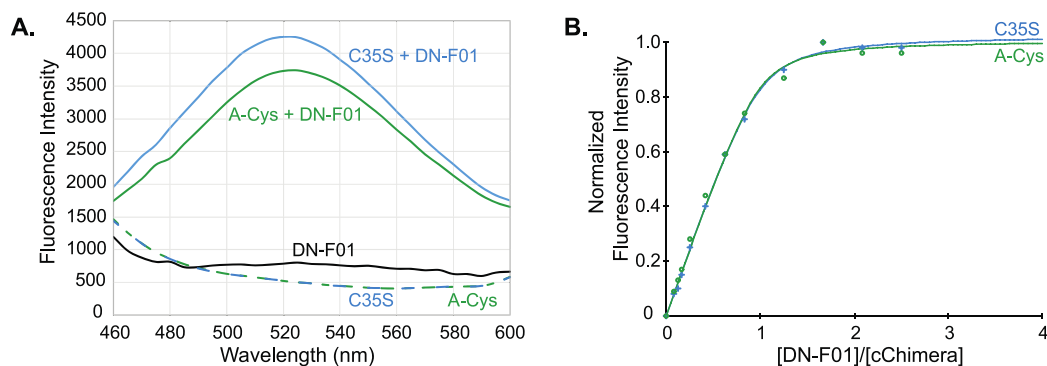


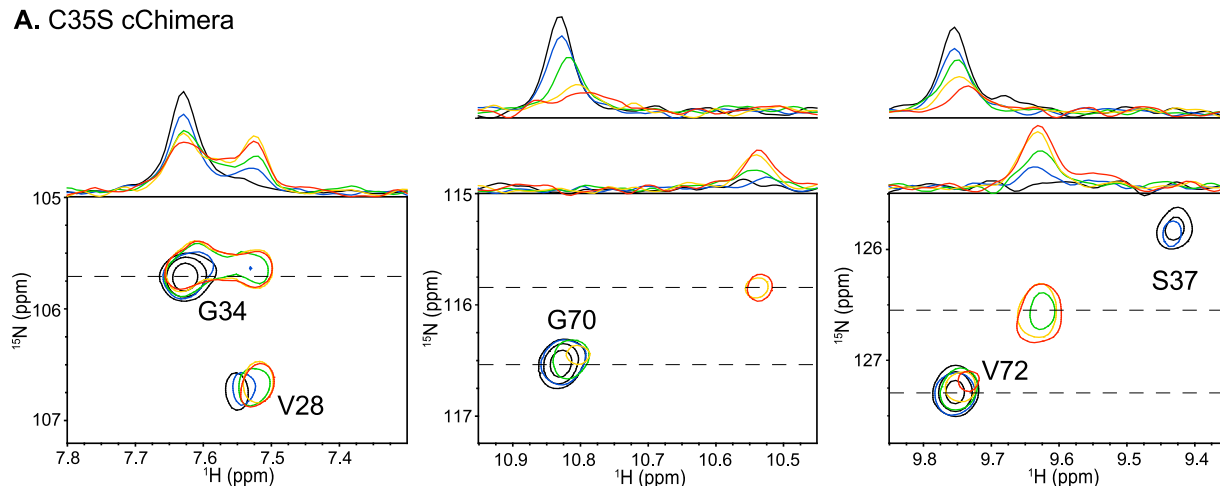
Figure 3. (A) Emission scans of 3 μ M DN-F01 (black), 1.2 μ M C35S cChimera (dashed blue), 1.2 μ M A-Cys cChimera (dashed green), 3 μ M DN-F01 + 1.2 μ M C35S cChimera (solid blue), and 3 μ M DN-F01 + 1.2 μ M A-Cys cChimera (solid green) in fluorescence buffer. (b) DN-F01 titrations to calcium saturated C35S (blue) and A-Cys (green) cChimera.

methyldiphenylamine (3-mDPA) binds deeply in the hydrophobic pocket of cNTnC and does not perturb the binding of cTnI switch region to cNTnC.²⁰ 3-Chlorodiphenylamine (3-CIDPA) was found to increase the calcium sensitivity of skinned cardiac muscle.⁹ DN-F01 also contains a nitrile group like levosimendan. It was reported that the nitrile group of DN-F01 reversibly covalently binds to cNTnC through a thioimide bond with Cys 84 of cNTnC, as does levosimendan.¹²

To assess the function of DN-F01 as a cardiac sarcomere modulator, actomyosin ATPase activities were measured using bovine cardiac myofibrils (CMFs). Although DN-F01 is a fluorescence dye, it does not interfere with Malachite green

assay for determining phosphate concentrations (Figure S1). Addition of DN-F01 to CMFs caused a drastic decrease in ATPase activity with an IC_{50} value (half maximal inhibitory concentration) of 11 ± 4 nmol/L (Figure 2A). DN-F01 had a strong inhibitory effect at both physiological calcium concentrations close to the values observed during peak systole (pCa 6) and supraphysiological calcium concentrations (pCa 4.5), but it showed no effect on the resting ATPase activity of CMFs in the absence of calcium (pCa 9) (Figure 2B). This suggests that DN-F01 specifically inhibits the calcium-dependent activation of the myofilaments. None of the other compounds studied previously, including the

A. C35S cChimera



B. A-Cys cChimera

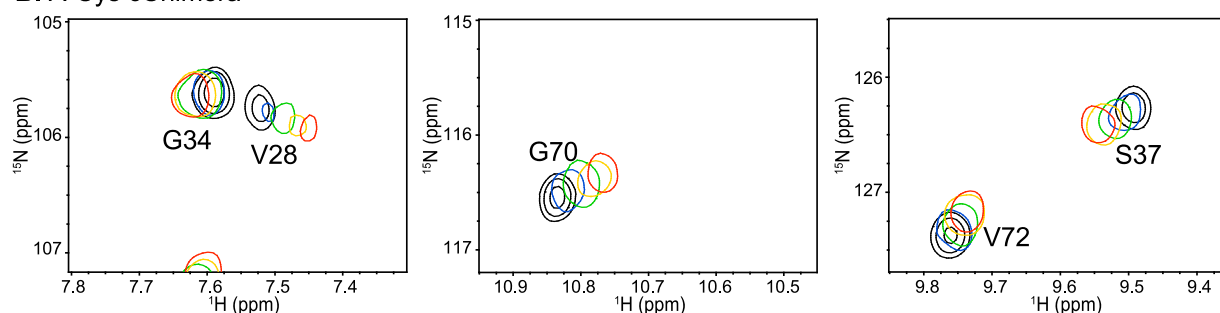


Figure 4. Regions of the 2D ^1H , ^{15}N HSQC NMR spectra of (A) C35S and (B) A-Cys cChimera during DN-F01 titrations. The first point in each titration is represented with multiple contours and subsequent titration points are represented by a single contour. For (A), [C35S cChimera] $\sim 40\ \mu\text{M}$ and [DN-F01] ~ 0 (black), 5 (blue), 15 (green), 25 (yellow), and 40 (red); the 1D NMR plots shown above the 2D contour plots the ^1H spectra at the ^{15}N frequencies indicated by dashed lines. For (B), [A-Cys cChimera] $\sim 25\ \mu\text{M}$ and [DN-F01] ~ 0 (black), 5 (blue), 15 (green), 30 (yellow), and 50 (red) μM .

reversible covalent compounds levosimendan and DLEV, showed a large effect on CMF ATPase.¹²

Covalent attachment of DN-F01 to cNTnC suggests that DN-F01 acts as an inhibitor by binding to the interface of cNTnC and cTnI.¹² In this study, we used the cNTnC-cTnI chimera (cChimera) to investigate the binding properties of DN-F01 to the interface of cNTnC and cTnI switch region. Several constructs of cChimera have been used for small molecules binding, protein dynamics and structure studies.^{20–22} The cChimera we used for this study incorporates cNTnC (residues 1–90), cTnI (residues 136–163), and a histidine tag. Residues 147–163 of cTnI are the switch region, and residues 136–146 of cTnI are used as a linker between cNTnC and cTnI switch region. The NMR solution structure indicates that cChimera reproduces the native interaction between the cNTnC and cTnI switch regions.²⁰ Although the wild type cNTnC has two cysteines (C35 and C84), a previous study found that DN-F01 is mainly covalently linked to cNTnC Cys 84.¹² To investigate the importance of Cys 84, we used two cChimera constructs: C35S cChimera and A-Cys cChimera. The C35S cChimera has only one cysteine, Cys 84, whereas the A-Cys cChimera has both C35S and C84S mutations.

Since DN-F01 is a fluorescent dye, we used fluorescence spectroscopy to study the binding property of DN-F01 to cChimera. Fluorescence samples were prepared in a 50 mM MOPS, 100 mM KCl, 3 mM MgCl_2 , 2 mM EGTA, and 2 mM CaCl_2 , pH 7.0 aqueous buffer, and all steady-state fluorescence

measurements were performed at 23 °C. Samples were excited at 400 nm, and emission spectra are shown in Figure 3A. Neither A-Cys nor C35S cChimera showed any fluorescence without the presence of DN-F01. DN-F01 alone showed a broad and weak fluorescence peak around 510–540 nm. The fluorescence intensity of DN-F01 drastically increased in the presence of cChimera, with a maximum at 520 nm for both C35S and A-Cys cChimera. To determine the binding affinity of DN-F01 to cChimera, we titrated 0 to 3 μM DN-F01 into 1.2 μM calcium-saturated cChimera and monitored the change in fluorescence intensity. The dissociation constants, K_D , of DN-F01 to C35S and A-Cys cChimera were 56 ± 16 and 45 ± 24 nM, respectively (Figure 3B). The binding constants of DN-F01 to both cChimera were in the same range with the IC_{50} value of the ATPase activity studies (considering different buffers and protein conditions), suggesting that the inhibition effect of DN-F01 results from binding to the cNTnC-cTnI interface and inhibiting the troponin complex. We noticed that there was no significant difference between the K_D of DN-F01 to A-Cys and the K_D of DN-F01 to C35S cChimera.

We also used NMR spectroscopy to further investigate how DN-F01 covalently binds. All NMR spectroscopic experiments used ^{15}N labeled A-Cys or C35S cChimera, and all data were collected at 30 °C on a Varian 600 MHz NMR spectrometer. The samples were prepared in aqueous buffer comprising 100 mM KCl, 10 mM imidazole, and 2 mM CaCl_2 at pH 6.7–6.8. We measured the solubility of DN-F01 in aqueous solution using both fluorescence and 1D ^1H NMR spectroscopies

(Figure S2), showing that the solubility of DN-F01 is less than 50 μM . The protein concentration used for DN-F01 titrations needs to be lower than the solubility limit of DN-F01 (less than 50 μM), which is challenging for 2D NMR spectroscopy. A protein concentration of 30 μM was used for A-Cys cChimera and 40 μM for C35S cChimera. The 2D NMR correlation spectrum at each titration point was acquired for 24 h (256 scans).

We monitored the titration of DN-F01 into C35S and A-Cys ^{15}N cChimera using 1D ^1H and 2D ^{15}N , ^1H HSQC correlation NMR. When DN-F01 was titrated into C35S cChimera, separate unliganded and liganded protein resonances were observed (Figure 4A), indicating slow exchange on the NMR time scale. Slow exchange happens when the k_{off} of protein–ligand complex is smaller than the difference between the chemical shifts of free and bound forms, so the signals from both free and bound states are observed. A slower k_{off} is typically associated with tighter binding. As the concentration of DN-F01 increased, peaks corresponding to the drug-free form decreased in intensity, while peaks corresponding to the drug-bound form increased in intensity (Figure 4A). This trend continued with the addition of DN-F01 until the concentration reached 40 μM , corresponding to a 1:1 ratio of DN-F01 to C35S cChimera concentration. The estimated dissociation constant for DN-F01 to C35S cChimera would be in the sub-micromolar range (typically $K_{\text{D}} < 0.5 \mu\text{M}$ for slow exchange), in good agreement with the K_{D} estimated by fluorescence titration experiments. The mass spectra of sample containing cChimera and DN-F01 solution showed two protein species: cChimera and cChimera-DN-F01 complex (Figure S3), further confirming the covalent binding of DN-F01 to C35S cChimera. As the concentration of DN-F01 was further increased, peaks from unliganded form started to shift, possibly due to secondary binding. On the other hand, when DN-F01 was titrated into A-Cys cChimera, the exchange of the DN-F01 was in the NMR fast exchange limit and the position of the protein resonances were the weighted average of that for the two forms (Figure 4B).

The reducing agent β -mercaptoethanol (BME) was used during the protein purification of C35S cChimera, and it was found later that BME was covalently linked to Cys 84 (by mass spectroscopy). When we titrated DN-F01 into C35S cChimera-BME, fast exchange was observed, similar to the DN-F01 titration into A-Cys cChimera. Cys 84 is covalently attached to BME, preventing it to form a covalent bond with DN-F01. However, when DN-F01 was titrated into C35S cChimera-BME in the presence of dithiothreitol (DTT), slow exchange was observed, similar to the DN-F01 titration into C35S cChimera (Figure S4). Cys 84 is reduced by DTT (confirmed by mass spectroscopy), and available to form a covalent bond with DN-F01. This further demonstrates that the slow exchange is observed because of the covalent binding between DN-F01 and Cys 84.

It has been shown that the reducing agent DTT interacts with levosimendan, possibly by reacting with the nitrile group.²³ As DN-F01 also containing the nitrile group, it might react with DTT. The stability of DN-F01 was monitored by 1D ^1H NMR, showing the aromatic region of DN-F01 in Figure S5. A freshly prepared DN-F01 was stable after 40 h at 30 $^{\circ}\text{C}$ in the absence of DTT. The spectrum changed immediately after adding excess DTT, indicating the formation of a new compound, but did not change significantly for the next 24 h.

We also observed that DN-F01 changed chemical structure over time. DN-F01 is a fluorophore, which may break down or degrade with exposure to light or changes in temperature, leading to irreversible photobleaching. The sample containing C35S cChimera and freshly made DN-F01 stock showed two protein species in the mass spectrum: cChimera and cChimera-DN-F01 (Figure S3). In contrast, a sample containing C35S cChimera and 8 month old DN-F01 stock showed more than two species. The most abundant new species were cChimera-DN-F01 + 202 Da and cChimera-DN-F01 + 404 Da (Figure S6). DN-F01 did not change structure in a short period of time (Figure S6), but it broke down and interacted with each other over long period of time.

Discovering and developing small molecules that target the cardiac sarcomere has been a great challenge over the past few decades, especially for sarcomere-based inhibitors. In this Note, we report a potent cardiac sarcomere inhibitor, DN-F01, targeting the thin filament protein troponin. ATPase activity studies show that DN-F01 has a strong inhibitory effect on cardiac myofilament calcium activation with nanomolar IC_{50} . Moreover, we show that DN-F01 covalently binds to Cys 84 on cTnI-cTnI chimera, and it also binds tightly non-covalently with K_{D} in the low nanomolar range. DN-F01 would be a useful starting point for further development of novel sarcomere-directed small molecule effectors for the treatment of hypertrophic cardiomyopathy. DN-F01 should also be an excellent tracer for screening of small molecule effectors that bind cTnI using fluorescence spectroscopy.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsmchemlett.1c00366>.

Experimental procedures for material, protein purification, ATPase activity experiments, fluorescence spectroscopy, nuclear magnetic resonance spectroscopy, and mass spectrometry (PDF)

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Notes

The authors declare no competing financial interest.

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